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CASE STUDY

Real-Time Visibility

Across Multi-State Operations

in Automotive Manufacturing

5+
Plants

2
Warehouses

100+
Suppliers

Tier-1 Automotive Component Manufacturer | Pan-India Operations | MES + ERP Integration

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Background

COMPANY PROFILE

A leading Tier-1 automotive component manufacturer supplying critical parts to OEMs, with operations spanning multiple states across India.

5+ Manufacturing Plants

2 Centralized Warehouses

100+ Suppliers Across Regions

THE CHALLENGE

As the company expanded geographically, major operational inefficiencies emerged — threatening delivery schedules, quality standards, and customer relationships.

- No real-time production visibility across plants
- Delayed decisions from fragmented data systems
- Inconsistent quality reporting between locations
- No centralized dashboard for management

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Business Challenges

No Real-Time Visibility

Lack of production visibility across geographically dispersed plants

Fragmented Data Systems

Delayed decision-making due to siloed, incompatible data systems

Inconsistent Quality

Inconsistent quality reporting and inspection standards between locations

WIP Tracking Gaps

Difficulty tracking Work-in-Progress inventory across plants

No Central Dashboard

Management had no unified view for monitoring overall performance

Poor Cross-Team Coordination

Siloed production, quality, and supply chain teams with no shared data

These challenges led to: Increased Downtime • Missed Delivery Schedules • Higher Rejection Rates • Reduced Customer Satisfaction

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Solution Implemented

Centralized Real-Time Manufacturing Visibility Platform (MES + ERP Integration)

01

Live Production Monitoring

Machine-wise & plant-wise real-time tracking dashboard

02

Multi-Plant Data Integration

Unified data from all states consolidated into one platform

03

WIP & Inventory Visibility

Real-time tracking across all plants and warehouses

04

Quality Management Integration

Standardized inspection and rejection tracking across sites

05

IoT Machine Connectivity

Automated data capture directly from shop floor machines

06

Centralized KPI Dashboard

OEE, downtime, and production efficiency at a glance

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Implementation Approach

01

Assessment Phase

- Evaluated existing IT systems and infrastructure
- Identified plant-level operational gaps
- Defined integration requirements for MES & ERP

02

Pilot Deployment

- Deployed platform in one plant first
- Validated system performance and accuracy
- Gathered feedback from operators & managers

03

Phased Rollout

- Gradual implementation across all plant locations
- Maintained operations continuity during rollout
- Real-time monitoring enabled from day one

04

Training & Change Mgmt

- Operator and management training sessions
- Standard operating procedures documented
- Ongoing support and system optimization

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Results & Impact

+25%	-30%	-20%	100%
Production Efficiency	Machine Downtime	Rejection Rates	Real-Time Visibility

Operational Improvements

- 25% increase in production efficiency
- 30% reduction in machine downtime
- 20% reduction in rejection rates

Visibility & Control

- 100% real-time visibility across all plants
- Faster decision-making with central dashboards
- Single source of truth for all operations

Supply Chain Optimization

- Improved inventory accuracy by 35%
- Better coordination between plants & warehouses
- Standardized processes across locations

Customer Satisfaction

- On-time delivery improved by 28%
- Stronger OEM relationships
- Data-driven delivery commitments

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Key Benefits & Conclusion

Single Source of Truth

All plants, teams, and management share one unified data view — no more silos

Data-Driven Decisions

Real-time KPIs empower faster, smarter decisions at every level of the organization

Standardized Processes

Uniform workflows, quality checks, and reporting across all plant locations

Future-Ready & Scalable

The platform is architected to support new plants, lines, and IoT integrations easily

Conclusion

By implementing a Real-Time Visibility Platform, the organization transformed from reactive operations to proactive, data-driven manufacturing.

The ability to monitor and control multi-state operations in real time became a critical competitive advantage in the automotive supply chain.

Reactive → Proactive | Fragmented → Unified | Data-Driven Manufacturing